# 2026-2027 – Econ 0107 – Macroeconomics I

## Chapter 6 : Welfare

(Chapter 6 in Azzimonti, Krusell, McKay, Mukoyama and others's textbook)

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# Outline

1. The First Welfare Theorem
2. Tracing Out the Pareto Frontier
3. Inefficient Market Outcomes
4. Overlapping Generations: Efficiency
5. Optimal Government Policy
6. Summary

# 1. The First Welfare Theorem

# Motivation

- Chapter 5 showed examples where the competitive equilibrium coincides with the planner's solution. The **First Welfare Theorem** (FWT) establishes this as a general result.
- Under conditions of the FWT, decentralized decisions by self-interested agents, coordinated only through prices, produce a Pareto-optimal outcome. This is Adam Smith's invisible hand.
- Pareto optimality is a minimal welfare criterion: silent on distribution. An allocation with extreme inequality can still be Pareto optimal.
- When the FWT fails – due to taxes, externalities, missing markets, or market power – understanding the nature of the failure is essential for evaluating policy.

# Abstract Setup

- Finite set $I$ of consumers, finite set $J$ of firms. Vectors of goods $x$, $y$, $\omega$.
- Resource constraint: $\sum_{i \in I} x_i \leq \sum_{j \in J} y_j + \sum_{i \in I} \omega_i$
- Notation:
  - $X_i$: consumption possibility set for consumer $i$
  - $Y_j$: production possibility set for firm $j$
  - $\succsim_i$: preference ordering for consumer $i$
  - $\theta_{i,j}$: consumer $i$'s ownership share of firm $j$, with $\sum_i \theta_{i,j} = 1$ for all $j$
  - $p$: price vector
- **Example**: static model, 3 goods (consumption good, capital, labour), one consumer with endowments of capital and labour $(k, 1)$, one firm with CRS production $y \leq F(k, \ell)$.

  $$x = (c, 0, 0) \in \mathbb{R}^3_+, \quad y = (y, -k, -\ell) \in \mathbb{R}_+ \times \mathbb{R}^2_-, \quad \omega = (0, k, 1), \quad p = (1, r, w)$$

- **Key assumption**: Local non-satiation (LNS) – each consumer can always be made better off by an infinitesimal change in consumption.

# Competitive Equilibrium (Abstract)

- A competitive equilibrium is a consumption allocation $\{x_i^\star\}_{\forall i}$, a production allocation $\{y_j^\star\}_{\forall j}$, and a price system $p^\star$ such that
  1. for each $i \in I$, $x_i^\star$ is in $X_i$ and there is no $x \in X_i$ such that $x \succ_i x_i^\star$ and

$$px \leq px_i^\star = p\omega_i + \sum_j \theta_{i,j} py_j$$

  2. for each $j \in J$, $y_j^\star \in Y_j$ and there is no $y \in Y_j$ such that $py > py_j^\star$
  3. market clearing: the resource constraint holds with equality

# Theorem 6.1 (First Welfare Theorem)

## Theorem 1

*An allocation that is part of a competitive equilibrium is Pareto optimal.*

**Proof** (by contradiction):

- Suppose $\{\tilde{x}_i\}_{\forall i}$, $\{\tilde{y}_j\}_{\forall j}$ is feasible and Pareto dominates the equilibrium:

$$\forall i \in I : \tilde{x}_i \succsim_i x_i^\star, \qquad \exists \tilde{\imath} \in I : \tilde{x}_{\tilde{\imath}} \succ_{\tilde{\imath}} x_{\tilde{\imath}}^\star$$

- From consumer optimization + LNS: $p\tilde{x}_i \geq px_i^\star$ for all $i$, with strict inequality for $\tilde{\imath}$.
- Summing household budgets:

$$\sum_{i \in I} p\tilde{x}_i > p\omega + \sum_{j \in J} py_j^\star$$

# Theorem 6.1 (First Welfare Theorem): Proof (cont.)

- From profit maximization: $p\tilde{y}_j \leq py_j^\star$ for all $j$. Therefore:

$$\sum_{i \in I} p\tilde{x}_i > p\omega + \sum_{j \in J} p\tilde{y}_j$$

- But feasibility requires

$$\sum_{i} p\tilde{x}_i \leq p\omega + \sum_{j} p\tilde{y}_j$$

- Contradiction. ■

# Mapping to Macroeconomic Models

- **Static model**: One consumer with endowments $(k, 1)$, one firm with CRS production.

$$x \in \mathbb{R}^3_+, \quad y = (y, -k, -\ell) \in \mathbb{R}_+ \times \mathbb{R}^2_-, \quad \omega = (0, k, 1), \quad p = (1, r, w)$$

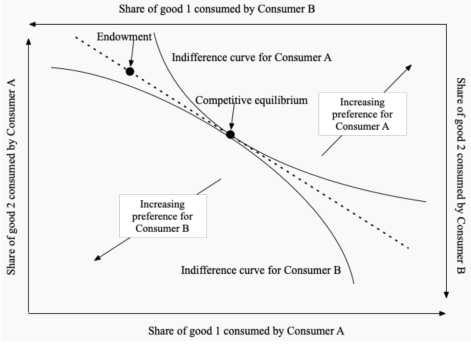
- **Dynamic model with** $T < \infty$: Goods at different dates are additional market goods. Vectors become $T$ times longer.
- **Infinite horizon**: Vectors are infinite-dimensional. The proof carries through provided total expenditures are finite:

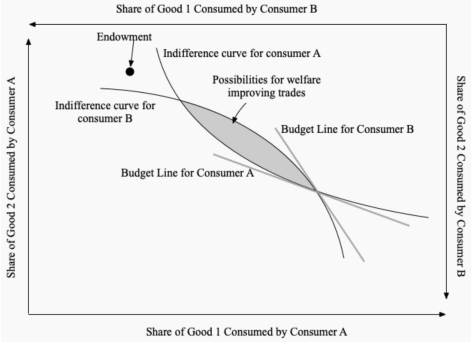
$$\sum_{t=0}^{\infty} p_t c_{i,t} < \infty$$

  × Typically satisfied in dynastic models ($p_t$ falls at rate $\beta < 1$). Can fail in OLG models.

# Intuition: Edgeworth Box

*Endowment economy with two goods and two consumers*




- **Panel (a)** (efficient equilibrium): Both consumers face the same prices. Indifference curves tangent to the budget line and to each other $\Rightarrow$ no Pareto improvement possible.
- **Panel (b)** (distorted market): Consumers face different (after-tax) prices. Indifference curves intersect $\Rightarrow$ shaded region represents Pareto-improving reallocations.

# 2. Tracing Out the Pareto Frontier

# The Social Planner's Problem

- A social planner maximizes a weighted sum of utilities:

$$\max_{\{c_{i,t}\}} \sum_{i \in I} \mu_i U_i, \qquad U_i = \sum_{t=0}^{\infty} \beta^t u(c_{i,t})$$

  subject to the resource constraint $\sum_{i \in I} c_{i,t} \leq \sum_{i \in I} \omega_{i,t}$ for all $t$.

- The weights $\mu_i$ are called **Negishi weights** (or Pareto weights).

- Any solution to this problem must be Pareto optimal: if it were not, the planner could increase the objective.

# Negishi Weights and Competitive Equilibrium

- Planner's FOC, where $\psi_t$ is the multiplier on the resource constraint at $t$:

$$c_{i,t} = (u')^{-1}\left(\beta^{-t}\frac{1}{\mu_i}\psi_t\right)$$

- Competitive equilibrium FOC (from Chapter 5):

$$c_{i,t} = (u')^{-1}(\beta^{-t}\lambda_i p_t)$$

- Setting $\mu_i = 1/\lambda_i$ and $\psi_t = p_t$: the two allocations coincide.
- **Implication**: The CE is optimal for a planner who assigns higher weight to agents with higher date-0 wealth ($\lambda_i$ lower $\Rightarrow$ $\mu_i$ higher).
- By varying $\{\mu_i\}$, we trace the **Pareto frontier**. The **Second Welfare Theorem** gives conditions (convexity) under which any Pareto-optimal allocation can be supported as a CE with appropriate lump-sum transfers.

# 3. Inefficient Market Outcomes

# Taxes

- **Lump-sum taxes**: Do not affect any marginal conditions $\Rightarrow$ act as redistribution along the Pareto frontier. FWT holds.
- **Proportional labor tax** $\tau_\ell$: With valued leisure, the after-tax wage is $w_t(1-\tau_\ell)$. MRS between consumption and leisure $\neq$ MRT $\Rightarrow$ FWT fails.
  - $\times$ Exception: If leisure is not valued (inelastic labor), the tax is non-distortionary
- **Capital income tax** $\tau_k$: After-tax return $r_t(1-\tau_k)$ enters the Euler equation. MRS between $t$ and $t+1$ goods $\neq$ MRT $\Rightarrow$ FWT fails.
- In both cases, consumers and firms face different price vectors ($p^c \neq p^f$) $\Rightarrow$ the consumer inequality (at $p^c$) and the profit inequality (at $p^f$) can no longer be chained: the contradiction in the proof disappears (see next slide).

# Where the Proof Breaks with a Proportional Labour Tax

- Firms face $p^f = (1, r, w)$, consumers face $p^c = (1, r, (1-\tau_\ell)w)$.
- Revenue on net trades is rebated lump-sum:

$$R = (p^c - p^f) \cdot \sum_i (x_i - \omega_i) = \sum_i T_i$$

- Consumer budget: $p^c x_i \le p^c \omega_i + \sum_j \theta_{i,j} p^f y_j^\star + T_i$.
- Suppose $\{\tilde{x}_i\}, \{\tilde{y}_j\}$ is feasible and Pareto dominates the equilibrium.
- **Step 1 (consumers + LNS)**: $p^c \tilde{x}_i \ge p^c x_i^\star$, strict for $\tilde{\imath}$. Summing, and using $R^\star = (p^c - p^f) \cdot \sum_j y_j^\star$ (market clearing):

$$\sum_i p^c \tilde{x}_i > p^c \omega + \sum_j p^f y_j^\star + R^\star = p^c \omega + \sum_j p^c y_j^\star \qquad \text{still holds (at } p^c\text{)}$$

# Where the Proof Breaks with a Proportional Labour Tax (cont.)

- **Step 2 (firms)**: we would need $p^c \tilde{y}_j \le p^c y_j^\star$, but firms maximize at $p^f$:

$$p^c \tilde{y}_j - p^c y_j^\star = \underbrace{p^f(\tilde{y}_j - y_j^\star)}_{\le 0} + \underbrace{(p^c - p^f)(\tilde{y}_j - y_j^\star)}_{\text{sign unknown}}$$

  × With a labour tax, the second term equals $\tau_\ell w(\tilde{\ell}_j - \ell_j^\star)$: positive whenever the alternative uses more labour, exactly the direction of a Pareto improvement.

- **Step 3 (feasibility)**: holds at any prices:

$$\sum_i p^c \tilde{x}_i \le p^c \omega + \sum_j p^c \tilde{y}_j$$

- Combined with Step 1 it only says $\sum_j p^c \tilde{y}_j > \sum_j p^c y_j^\star$: no contradiction.

# Externalities: Competitive Equilibrium

- An externality arises when one agent's activity has payoff relevance to others, not reflected in prices.
- **Example**: Firm $j$ produces $A(\bar{y})k_j^{\alpha}\ell_j^{1-\alpha}$, where $A$ is decreasing in average output $\bar{y}$.
  - $\times$ Continuum of firms on $[0,1]$: each firm takes $\bar{y}$ as given.
- **Consumer**:

$$\max_{c,\ell} \ \log c - B\frac{\ell^{1+1/\theta}}{1+1/\theta} \quad \text{s.t.} \quad c = rk + w\ell$$

- FOC:

$$\frac{w}{rk + w\ell} = B\ell^{\frac{1}{\theta}}$$

# Externalities: Competitive Equilibrium (cont.)

- **Firm**: $\max_{k,\ell} A(\bar{y})k^{\alpha}\ell^{1-\alpha} - rk - w\ell$, taking $\bar{y}$ as given. FOCs:

$$r = \alpha A(\bar{y})k^{\alpha-1}\ell^{1-\alpha}, \quad w = (1-\alpha)A(\bar{y})k^{\alpha}\ell^{-\alpha} \quad \Rightarrow \quad \frac{r}{w} = \frac{\alpha}{1-\alpha}\frac{\ell}{k}$$

- **Combining**: $rk = \frac{\alpha}{1-\alpha}w\ell$, so income is $rk + w\ell = \frac{w\ell}{1-\alpha}$.
- Substituting in the consumer FOC:

$$\frac{(1-\alpha)w}{w\ell} = B\ell^{\frac{1}{\theta}} \quad \Rightarrow \quad \boxed{1-\alpha = B\ell^{1+\frac{1}{\theta}}}$$

- Equilibrium hours do not depend on $A$ (income and substitution effects cancel).

# Externalities: Efficient Allocation

- **Planner** (internalizes the effect of output on TFP):

$$\max_{\ell,y} \ \log y - B\frac{\ell^{1+1/\theta}}{1+1/\theta} \quad \text{s.t.} \quad y = A(y)k^{\alpha}\ell^{1-\alpha} \qquad (\text{multiplier } \lambda)$$

- FOCs:

$$\ell: \ B\ell^{\frac{1}{\theta}} = \lambda(1-\alpha)A(y)k^{\alpha}\ell^{-\alpha}, \qquad y: \ \frac{1}{y} = \lambda\left(1 - A'(y)k^{\alpha}\ell^{1-\alpha}\right)$$

- Multiply the first by $\ell$ and use the constraint: $B\ell^{1+\frac{1}{\theta}} = \lambda(1-\alpha)y$.
- The second gives $\lambda y = 1/\left(1 - A'(y)k^{\alpha}\ell^{1-\alpha}\right)$. Hence

$$\boxed{\frac{1-\alpha}{1 - A'(y)k^{\alpha}\ell^{1-\alpha}} = B\ell^{1+\frac{1}{\theta}}}, \qquad y = A(y)k^{\alpha}\ell^{1-\alpha}$$

# Externalities: Why the Equilibrium Is Inefficient

| Equilibrium | Planner |
|---|---|
| $1-\alpha = B\ell^{1+\frac{1}{\theta}}$ | $\frac{1-\alpha}{1-A'(y)k^{\alpha}\ell^{1-\alpha}} = B\ell^{1+\frac{1}{\theta}}$ |

- **Comparison**: with $A' < 0$ the denominator exceeds $1 \Rightarrow$ efficient hours are *lower* than equilibrium hours.

- The equilibrium is inefficient unless $A'(y) = 0$ (no externality
- In equilibrium, firms ignore how their production hurts others.

- **What goes wrong in the proof**:
  - $\times$ Production set $Y_j = Y_j\big((y_{j'})_{j' \neq j}\big)$ is endogenous and interdependent.
  - $\times$ The step $p\tilde{y}_j \leq py_j^{\star}$ no longer follows: $\tilde{y}_j$ need not belong to the $Y_j$ the firm faced in equilibrium.

# Missing Markets: Borrowing Constraints

- Two-agent dynastic endowment economy, log utility.
- Endowments alternate: agent 1 has $2\omega/3$ in odd periods, $\omega/3$ in even periods; agent 2 reversed.
- **Unrestricted equilibrium**: Full consumption smoothing, $c_{1,t} = c_1$ and $c_{2,t} = c_2$ for all $t$, MRS$=\beta u'(c_{t+1})/u'(c_t) = q_t \rightsquigarrow$ Gross interest rate $= 1/\beta$.
- **With borrowing constraint** $a_{t+1} \geq 0$: Euler equation becomes an inequality when the borrowing constraint binds:

$$q_t u'(c_t) \geq \beta u'(c_{t+1})$$

- **Result**: autarky. No smoothing. Clearly not Pareto optimal.
- Equilibrium bond price from the unconstrained agent Euler equation $q_t = 2\beta$ (gross interest rate $= 1/(2\beta)$, can be negative if $\beta > 0.5$).
  - $\times$ The unconstrained agent sets the price (meaning their MRS is equal to relative price); the constrained agent would like to borrow but cannot.

# Borrowing Constraints: Who Sets the Interest Rate?

- Budget $c_t + q_t a_{t+1} = \omega_t + a_t$, constraint $a_{t+1} \geq 0$. FOC for $a_{t+1}$, with Kuhn-Tucker multiplier $\mu_t \geq 0$:

$$\underbrace{q_t}_{\text{rel. price}} = \underbrace{\frac{\beta c_t}{c_{t+1}}}_{\text{MRS}} + \mu_t c_t$$

- Even $t$, autarky: agent 1 is poor today ($\omega/3$), rich tomorrow ($2\omega/3$); agent 2 the reverse.
- Bond market clearing with $a_{t+1} \geq 0$ for both $\Rightarrow a_{t+1} = 0$: nobody borrows, so nobody can lend.
- Which Euler equation holds with equality determines $q_t$.

# Borrowing Constraints: Who Sets the Interest Rate? (cont.)

- **Agent 2 (would-be lender) is unconstrained**: $a_{t+1} = 0$ is an interior choice $\Rightarrow$ $\mu_t = 0$:

$$q_t = \frac{\beta 2\omega/3}{\omega/3} \quad \Rightarrow \quad q_t = 2\beta \qquad \text{MRS} = 2\beta$$

  - $\times$ The bond price must make agent 2 *just willing* not to lend: the interest rate falls until lending is discouraged.

- **Agent 1 (would-be borrower) is constrained**: at $q_t = 2\beta$,

$$q_t = \frac{\beta\omega/3}{2\omega/3} + \mu_t\omega/3 \quad \Rightarrow \quad \mu_t = 3/2 > 0 \qquad \text{MRS} = \beta/2$$

  - $\times$ Marginal value of consumption today exceeds its market cost in terms of tomorrow: agent 1 wants to borrow, but $a_{t+1} \geq 0$ binds. Its Euler inequality does not pin down $q_t$.

- MRS differ across agents ($2\beta$ vs. $\beta/2$) $\Rightarrow$ not Pareto optimal.

# Borrowing Constraints: Pecuniary Externalities

- **What goes wrong in the proof?** As with externalities, the possibility sets $X_i$ change:
  - × They contain *more restrictions*: one extra constraint per good, e.g. $c_0 \leq a_0 + \omega_0$, $q_0 c_1 \leq \omega_0 + a_0 - c_0 - q_0 \omega_1$, ...
  - × The equilibrium can be dominated by an allocation (e.g. full smoothing) that is simply *not in* the agent's consumption possibility set.
- The sets are also **endogenous**: they depend on prices ($q_t$ appears in the constraints).
  - × Departs from the abstract setup, where $X_i$ is exogenous.
  - × Interdependence: through prices, each agent's choices shape the sets the others face.
- **Pecuniary externality**: one agent's choices affect the prices that appear in other agents' constraints, and that agent does not take this into account.
- A planner choosing allocations subject to the same budget and borrowing constraints would internalize these effects $\Rightarrow$ may choose a different allocation than the competitive equilibrium.

# Pecuniary Externalities

- We refer to *pecuniary externalities* when one agent's choices affect the prices that appear in other agents' constraints, and that agent does not take this into account.
- It is a bit of a misleading concept,
- Think of competitive equilibrium without frictions
  - × If agent 1 suddenly likes a lot apples, the equilibrium price of apples will go up, hurting the other apple consumers. But this is not internalized by the agent (pecuniary externalities)
  - × Nevertheless, competitive allocations are optimal because the increase in price will exactly make agent 1 internalize what needs to be internalized (the invisible hand).
- Whether pecuniary externalities will make the equilibrium inefficient must be decided on a case by case basis.

# Lack of Commitment

- When agents can default on debt without punishment: leads to de-facto borrowing constraints (rational lenders refuse to lend).
- With punishment: intertemporal trade occurs, but the punishment mechanism itself must be committed to.
- **Examples studied later**:
  - × Government cannot commit to future tax policy (Chapter 15)
  - × Sovereign debt default (Chapter 24)

# Market Power

- Some agents have **market power**: through their own behavior, they can affect the prices they transact at.
  - × Individual level: a worker may bargain for a higher salary (labor-market frictions, later chapters).
  - × Firm level: a firm sets the price of its product above marginal cost (**markups**), or bargains with input suppliers (growth and business cycles, later chapters).
- In macro models, market power is almost always assumed to be **limited**: it affects an individual transaction, not aggregates.
  - × **Monopolistic competition** (Dixit and Stiglitz, 1977): each firm has a monopoly over a specific good but competes with a continuum of imperfect substitutes, each negligible in the aggregate.
  - × Likewise, a worker has no influence on aggregates in wage negotiations.
  - × Exceptions exist (Apple is *granular* and may affect the whole smartphone market), but are likely rare.
- **FWT**: the abstract proof cannot be used directly, since some agents do not take $p$ as given.

# Monopolistic Competition: Setup

- Consumer with CES preferences over a continuum of goods, where $\varepsilon > 1$ is the elasticity of substitution:

$$U\Big(\{c(i)\}_{i=0}^{1}, L\Big) = u\left(\left(\int_0^1 c(i)^{1-\frac{1}{\varepsilon}}\, di\right)^{\frac{\varepsilon}{\varepsilon-1}}\right) - v(L)$$

- Consumer budget: $\displaystyle\int_0^1 p(i)c(i)\, di = y$
- **CES price index** (unit cost of composite $C$):

$$P = \left(\int_0^1 p(i)^{1-\varepsilon}\, di\right)^{\frac{1}{1-\varepsilon}}$$

- **Demand for variety** $i$: $c(i) = C\left(\dfrac{p(i)}{P}\right)^{-\varepsilon}$

# Monopolistic Competition: Equilibrium

- Each firm produces $c(i) = A\ell(i)$ and sets price facing the demand curve. Total labor supply $L = 1$ (inelastic).
- **Firm's problem**: $\max_{p,c,\ell} pc - w\ell$ subject to $c = A\ell$ and $p = P(c/C)^{-1/\varepsilon}$.
- **Optimal price** (constant markup $\mu = \varepsilon/(\varepsilon - 1) > 1$):

$$p = \frac{\varepsilon}{\varepsilon - 1} \cdot \frac{w}{A}$$

- **Equilibrium wage**: $w = A(1 - 1/\varepsilon)$ (below marginal product).
- **Equilibrium output** (with $u = \log$, $P = 1$): $A(1 - 1/\varepsilon) = Cv'(C/A)$
- **Planner's optimum**: $A = Cv'(C/A)$. Output and hours are **too low** in equilibrium.

# Quantifying Welfare Losses

- As an example, compare two allocations: $(c_l, \ell_l)$ under lump-sum taxes vs. $(c_d, \ell_d)$ under distortionary taxes.
- Define the **consumption-equivalent welfare loss** $\Delta > 0$:

$$u(c_l(1-\Delta), \ell_l) = u(c_d, \ell_d)$$

- $\Delta$ is the percentage reduction in consumption (at fixed hours) that makes the better allocation as good as the worse one.
  - × Has a real interpretation (unlike "utils")
  - × In dynamic models: $\Delta$ applies uniformly across all periods
  - × Alternative: $\Delta$ as a percentage *increase* in the $d$ allocation needed to reach the $l$ one (gives a different number)

# 4. Overlapping Generations: Efficiency

# Why Are OLG Models Interesting?

- **A striking welfare result**: even *without* frictions – no missing markets, distortionary taxes, monopoly power, or externalities – equilibria can be (though need not be) Pareto inefficient.
  - × And there is a simple litmus test to tell whether an equilibrium is efficient.
- **Why the FWT can fail**: infinite horizon *and* infinitely many finitely-lived agents. No single agent spans all dates, so markets cannot link all generations.
- **A realistic demographic structure**: finite lives, saving for retirement, heterogeneity across cohorts.
  - × Related models where people die randomly and new people appear share these properties.
- **Policy relevance**: when the equilibrium is inefficient, intergenerational transfers can make *everyone* better off.
  - × Pay-as-you-go pensions, government debt, fiat money (Samuelson, 1958; Diamond, 1965).
  - × Dynamic inefficiency (oversaving, interest rate below growth rate) and the debate on public debt when $r < g$.

# OLG Endowment Economy: Setup

- Representative consumer per cohort, two-period lives, $u_t(c_y, c_o) = \log c_y + \log c_o$.
- Old of generation -1: $u_{-1} = \log c$
- Stationary endowments $(\omega_y, \omega_o)$.

- Competitive equilibrium: sequential trading, no borrowing constraints.

- Consumer born at $t$ solves:

$$\max_{c_y, c_o} \log c_y + \log c_o \quad \text{s.t.} \quad c_y + q_t c_o = \omega_y + q_t \omega_o$$

- Generation $t$ consumer cannot do intertemporal trade (no other generation lives in $t$ and $t+1$).
- **Equilibrium**: autarky, $c_{y,t} = \omega_y$, $c_{o,t} = \omega_o$, bond price $q_t = \omega_y / \omega_o$ (from Euler equation).

# OLG Endowment Economy: Is the Equilibrium Efficient?

- **Case 1 ("incentives to save")**: $\omega_y = 3$, $\omega_o = 1$. Bond price $q = 3$ (gross interest rate 1/3, interest rate = -66%).
  - × Alternative allocation: $\tilde{c}_{y,t} = \tilde{c}_{o,t} = 2$ for all $t$ (transfer 1 from young to old each period) (it is feasible).
  - × Initial old: strictly better ($2 > 1$)
  - × All subsequent generations: $\log 2 + \log 2 = \log 4 > \log 3 + \log 1 = \log 3 \Rightarrow$ strictly better
  - × $\Rightarrow$ Equilibrium (autarky) is **not** Pareto optimal.

- **Case 2 ("incentives to borrow")**: $\omega_y = 1$, $\omega_o = 3$. Bond price $q = 1/3$ (gross interest rate 3, interest rate = +200%).
  - × The $(2, 2)$ alternative improves all generations born at $t \geq 0$ but makes the initial old worse off ($2 < 3$).
  - × No other allocation can Pareto dominate $\Rightarrow$ equilibrium (autarky) **is** Pareto optimal.

# Revisiting the Proof of the FWT in OLG

- In the OLG model, equilibria are sometimes Pareto optimal and sometimes not. Why?
- Go back to the abstract proof of the FWT: the notation can include infinite sequences, and the proof goes through in all parts, except possibly one:
  - × when summing the budgets over all agents, $I$ is now **infinite**.
- Prices are the sequence $p = \{p_0, p_1, p_2, \ldots\}$, with $p_t = \prod_{\tau=0}^{t-1} q_\tau$.
- $\omega_y = 3$, $\omega_o = 1$: $q_t = 3$ **for all** $t$: the present value of the equilibrium allocation is

$$\sum_{t=0}^{\infty} p_t(\omega_y + \omega_o) = (\omega_y + \omega_o)\left(1 + 3 + 3^2 + \cdots\right) = \infty$$

  - × This proof strategy cannot be used (indeed, the $(2, 2)$ allocation Pareto dominates the equilibrium).
- $\omega_y = 1$, $\omega_o = 3$: $q_t = 1/3$ **for all** $t$: the present value is finite, and the proof does go through.
  - × We in fact have a proof that no allocation can Pareto improve on the autarkic allocation $(c_{y,t}, c_{o,t}) = (1, 3)$ for all $t$.

# The Balasko-Shell Theorem (Endowment Economy)

## Theorem 2 (Balasko and Shell, 1980)

*A competitive equilibrium in an OLG endowment economy is Pareto optimal if and only if*

$$\sum_{t=0}^{\infty} \frac{1}{p_t} = \infty$$

*where* $p_t = \prod_{\tau=0}^{t-1} q_\tau$ *is the price of consumption at* $t$ *(the inverse of the gross interest rate between 0 and* $t$*).*

- **Implication**: When the gross real interest rate converges, the equilibrium is efficient if and only if the limit net real interest rate is non-negative.
- With growth at rate $g$: efficient if and only if the limit interest rate $\geq g$.
- When the equilibrium is inefficient, a PAYG transfer scheme (young $\rightarrow$ old) makes all generations better off. But this requires government commitment to an infinite sequence of transfers.

# Dynamic Efficiency with Production

- Neoclassical production economy with OLG.
- $(\omega_y, \omega_o)$ is the endowments of a cohort in labor efficiency units.

$$c_y + s = \omega_y w_t, \qquad c_o = s(1 - \delta + r_{t+1}) + \omega_o w_{t+1}$$

- Asset market clearing: $s_t = k_{t+1}$. Competitive pricing: $r_t = F_1(k_t, \omega_y + \omega_o)$, $w_t = F_2(k_t, \omega_y + \omega_o)$.

# Dynamic (In)Efficiency

- **Dynamic (in)efficiency** is a *production* concept about the capital path $\{k_t\}$:
- Dynamically inefficient: another feasible path $\{k_t\}$ gives at least as much aggregate consumption at every date, strictly more at some date.
- In the standard Diamond OLG model: the equilibrium is Pareto optimal iff it is dynamically efficient.

Theorem 3

*Define* $R_t = 1 - \delta + F_1(k_t, \omega_t)$. *Then* $\{k_t\}_{t=0}^{\infty}$ *is dynamically efficient if and only if*

$$\sum_{t=0}^{\infty} \prod_{s=1}^{t} R_s(k_s) = \infty$$

- **Steady state**: efficient iff $R = F_1(k_{ss}, \omega) + 1 - \delta \geq 1$, i.e., the net interest rate is non-negative.

# Dynamic Inefficiency: Intuition

- **When $R < 1$ at steady state**:
  - × Reduce saving by $\varepsilon$ at $t$. Resources freed at $t$.
  - × At $t+1$, production falls by less than $\varepsilon$ (since marginal return $< 1$).
  - × Reduce saving by $\varepsilon$ again at $t+1$: net resources freed.
  - × Repeat forever $\Rightarrow$ **free lunch** at all dates.

- **When $R \geq 1$**:
  - × Reducing saving by $\varepsilon$ at $t$ reduces resources at $t+1$ by more than $\varepsilon$.
  - × Future reductions grow and eventually become infeasible.

# Dynamic Inefficiency: Oversaving

- **Oversaving**: In OLG models, equilibrium savings can be "too high" because each generation saves for retirement without regard for the aggregate capital stock.
- In such a case of dynamic inefficiency, rational bubbles can be sustained and they are Pareto improving. They are a mean of reducing investment in $k$.

- Fundamentally different from the dynastic model, where the TVC prevents over-accumulation.

- The warm-glow model (Chapter 5) can also generate dynamic inefficiency: when $v(b) = Ab$ (linear), convergence to a steady state with $r - \delta < 0$.

# 5. Optimal Government Policy

# Correcting Market Failures

- **Externalities** (Pigou, 1920): A tax equal to the marginal external damage, evaluated at the efficient allocation, restores optimality.
  - × Alternative: assign property rights (Coase) and let the market internalize the externality. Not always feasible.
- **Monopoly**: Subsidize production at a per-unit rate to counteract under-production. Alternatively, anti-trust regulation.
  - × But monopoly power can incentivize innovation through patents ⇒ trade-off (Chapter 13).
- **Distortionary taxes**: Compare alternative feasible tax systems (Ramsey analysis, Chapter 15). Which tax base is least distortionary?
- **Missing markets**: Beware the "chicken model" – if a market is missing, it is usually missing for a reason (moral hazard, adverse selection).
  - × Mirrlees (1971) approach: model the friction explicitly.

# Redistribution

- A separate aim from efficiency: achieving a more equitable distribution, even if some agents are made worse off.
- Common approach: utilitarian social welfare function with equal weights, where $\omega_i$ is the fraction of type-$i$ agents:

$$\max \sum_i \omega_i u(c_i) \quad \text{s.t.} \quad \sum_i \omega_i c_i = \sum_i \pi_i \omega_i$$

- With non-distortionary redistribution: equal weights $\Rightarrow$ equal consumption. With distortionary taxes: optimal redistribution is partial.
- **Behind the veil of ignorance**: Before knowing one's type, an agent would choose the utilitarian optimum as an insurance scheme.
  - $\times$ Cannot be offered by markets (agents not yet born), but government can implement it.

# Redistribution with Private Information: Mirrlees (1971)

- **The friction**: workers differ in productivity $w_i$, which is their *private information*. The government observes only income $y_i = w_i \ell_i$, not $w_i$ or $\ell_i$ separately.
  - $\times$ First best (lump-sum transfers conditioned on $w_i$) is not available: high types would claim to be low types.
- **Mechanism design problem**: choose bundles $\{(c_i, y_i)\}$ to solve

$$\max \sum_i \omega_i \left[ u(c_i) - v(y_i/w_i) \right] \quad \text{s.t.} \quad \sum_i \omega_i c_i = \sum_i \omega_i y_i$$

  and the **incentive-compatibility** constraints: for all $i, j$,

$$u(c_i) - v(y_i/w_i) \;\geq\; u(c_j) - v(y_j/w_i)$$

- **Results**: an equity-efficiency trade-off; optimal redistribution is partial and implemented by a (distortionary) nonlinear income tax.
- **Lesson** (vs. the "chicken model"): model the friction explicitly, so the government faces the same informational constraints as the market.

# 6. Summary

# Key Takeaways (I)

- **First Welfare Theorem**: Under LNS, a competitive equilibrium is Pareto optimal.
  - × Proof: the alternative allocation must be more expensive (consumer optimization) but cannot be (profit maximization + feasibility). Contradiction.
- **Negishi weights**: CE is optimal for a planner with $\mu_i = 1/\lambda_i$. Varying weights traces the Pareto frontier.
  - × Second Welfare Theorem: any Pareto optimum can be decentralized with lump-sum transfers (under convexity).
- **Market failures**:
  - × Distortionary taxes (consumers and firms face different prices)
  - × Externalities (possibility sets interdependent)
  - × Missing markets (additional constraints, pecuniary externalities)
  - × Lack of commitment
  - × Monopoly power (price-setting, not price-taking)

# Key Takeaways (II)

- **OLG efficiency**: Even without frictions, OLG equilibria can be Pareto inefficient.
  - × Balasko-Shell: efficient iff $\sum 1/p_t = \infty$, i.e., the asymptotic net interest rate $\geq 0$ (or $\geq g$ with growth).
  - × Dynamic inefficiency = oversaving.
- **Welfare measurement**: Consumption-equivalent $\Delta$ – the percentage reduction in consumption that equates utility across allocations.
- **CES and monopolistic competition**: Markup $\mu = \varepsilon/(\varepsilon - 1)$; CES price index $P = (\int p(i)^{1-\varepsilon} di)^{1/(1-\varepsilon)}$; equilibrium under-produces relative to planner.
- **Policy**: Pigou taxes for externalities, Ramsey analysis for tax design, Mirrlees for informational frictions. Beware the "chicken model."

